Requirements Table

Problem Statement #16 — Remote Patient Vitals Alert & Monitoring App | Lab 1: Requirements Engineering & UML Use-Case Modelling

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall continuously evaluate ingested patient vitals against configured clinical thresholds and flag critical anomalies within 2 seconds.	High	Pass: High heart rate (>140 BPM) triggers immediate escalation alert. Fail: Metric spike ignored or delayed > 5s.	Given (baseline requirement).
FR-002	Functional	The system shall escalate an unacknowledged critical alert to the next tier of the on-call caregiver matrix if it is not acknowledged within 3 minutes of being raised.	High	Pass: A simulated unacknowledged alert is automatically re-routed to Tier-2 caregiver at exactly 3 min, logged with timestamp. Fail: Alert remains only with the original caregiver beyond 3 min 30s.	Ensures a missed alert never goes unnoticed and reaches an available caregiver in time-critical post-operative situations.
FR-003	Functional	The system shall allow an On-Call Caregiver to acknowledge an active alert, recording the caregiver ID and acknowledgment timestamp against the alert.	High	Pass: Tapping "Acknowledge" stops further escalation and stores caregiver ID + timestamp in the alert log within 1s. Fail: Alert continues to escalate after acknowledgment, or acknowledgment is not logged.	Provides an auditable chain of custody for every emergency alert, required for clinical accountability.
FR-004	Functional	The system shall allow an authorized System Administrator to configure and update per-patient clinical thresholds for SpO2, Heart Rate, and Blood Pressure.	Medium	Pass: Updated threshold values are saved, version-stamped, and applied to the next incoming telemetry reading within 10s. Fail: Old threshold values are still used after an update, or the change is not saved.	Post-operative patients have individualized clinical baselines, so thresholds cannot be one-size-fits-all.
FR-005	Functional	The system shall automatically notify the Emergency Dispatch Service when an escalated alert remains unacknowledged after the final caregiver tier's timeout has elapsed.	High	Pass: If no caregiver acknowledges within the final tier's timeout, an emergency dispatch request with patient location and vitals snapshot is transmitted within 5s. Fail: No dispatch notification is sent, or it is sent before all caregiver tiers have timed out.	Acts as the last line of defense to guarantee patient safety when the human caregiver chain fails to respond.
NFR-001	Nonfunctional	The telemetry ingestion gateway shall support at least 500 concurrent continuous telemetry streams with 99.99% uptime.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Given (baseline requirement).
NFR-002	Nonfunctional	The system shall encrypt all patient vitals data, both in transit (TLS 1.2+) and at rest (AES-256), and shall comply with HIPAA data-protection requirements.	High	Pass: Security audit / penetration test confirms no vitals data is transmitted or stored unencrypted, and access logs show role-based access control. Fail: Any patient data field is found in plaintext in transit or at rest.	Vitals and location data are sensitive PHI; a breach directly threatens patient privacy and regulatory compliance (Type: Security & Compliance).

Note: FR-001 and NFR-001 were provided in the problem statement; FR-002–FR-005 and NFR-002 were authored for this lab.